

Questions on Deep Neural Networks

What is the building block of neural networks?

What is a perceptron?

Which problems can be solved by single layer perceptrons?

How can it be trained?

What are common loss functions?

What is different in ConvNets from plain neural networks?

What is First ConvNet? & First ConvNet on ImageNet Dataset?

What is different between the above two? What is similar?

What activation functions can we use?

Why pooling?

Importance of depth: What features are learned in lower, what in deeper layers?

Feature generalization:

What is fine-tuning?

Why/When do we need it?

What deeper architectures do you know?

What are possible issues when training deep architectures?

What different techniques have we seen that make training deep architectures more stable?

Describe the inception unit?

Describe the structure of the residual unit?